

Anomalous Aortic Origin of a Coronary Artery: Results from a Single Surgical Team in Spain

World Journal for Pediatric and
Congenital Heart Surgery
2025, Vol. 16(1) 64-72
© The Author(s) 2024
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/21501351241278684
journals.sagepub.com/home/pch



Laura Varela Barca, PhD¹ , Rafael Hernández-Estefanía, PhD², Miguel Orejas Orejas, PhD³, Alicia Donado Miñambres, MD², Marta Tomás Mallebrera, MD⁴, Pilar Calderón Romero, MD², Angeles Heredero Yung, PhD², and Gonzalo Aldámiz-Echevarría, MD²

Abstract

Objectives: Anomalous aortic origin of a coronary artery is a rare congenital lesion in which a coronary artery arises from an anomalous location within the aorta. Anomalous aortic origin of a coronary artery has been associated with myocardial ischemia and it is considered the second most common cause of sudden cardiac arrest in young athletes. When surgical repair is indicated, surgical unroofing is the most commonly employed technique. Our objective is to describe the outcomes of our surgically treated patients. **Methods:** We present a series of 16 adult patients who underwent surgical repair of anomalous aortic origin of a coronary artery. Patients were treated in three different institutions by the same surgeon. Surgical unroofing of the anomalous coronary artery was the surgical technique chosen in the majority of the patients. Follow-up was performed. **Results:** Unroofing of an intramural anomalous coronary artery was the procedure performed in 11 patients. Three patients underwent neo-ostium creation; one patient underwent a David procedure with coronary reimplantation; and one patient was treated with coronary bypass grafting due to severe coronary atheromatous lesions. There were no perioperative deaths, and no major postoperative complications. Follow-up period was 73.8 months, the survival rate was 100%, and there were neither ischemia or heart failure reports. **Conclusions:** The surgical repair of anomalous aortic origin of a coronary artery by coronary unroofing or neo-ostium creation has demonstrated excellent early and late outcomes. Late survival was excellent. The follow-up period revealed no significant morbidity or complications.

Keywords

anomalous aortic origin of a coronary artery, surgical coronary repair, long-term outcomes

Submitted April 12, 2024; Accepted August 9, 2024.

Introduction

Anomalous aortic origin of a coronary artery (AAOCA) is a rare congenital coronary anomaly described first in the Renaissance period, but it was not considered a clinically significant issue until the mid-1970s.¹ Anomalous aortic origin of a coronary artery occurs in approximately 0.2% to 1.33% of the general population² and it is defined as a congenital malformation in which a coronary artery arises from a place in the aorta other than the normal sinus of Valsalva.³ Anomalous aortic origin of a coronary artery includes a high variability of anatomical variants since the origin can be in the ascending aorta, the opposite sinus of Valsalva, or the noncoronary sinus. Multiple classifications of AAOCA have been proposed, based on anatomy, function, or clinical significance.⁴ In 1999, the nomenclature was standardized by the STS-Congenital Heart Surgery Database Committee and the European Association for Cardiothoracic Surgery.⁵ Anomalous aortic origin of a coronary artery can be subdivided into anomalous aortic origin of the left coronary artery (AAOLCA) and anomalous aortic origin of the right coronary artery (AAORCA).⁶ The relative incidence of

AAORCA is estimated to be 3 to 9 times more common than AAOLCA.⁷

The importance of AAOCA is due to its clinical significance: AAOCA has been associated with myocardial ischemia and it is considered the second most common cause of sudden cardiac arrest in young athletes.^{8,9} The risk of sudden death is widely acknowledged to be higher in AAOLCA than in AAORCA.⁶ The mechanisms suggested as possible causes for sudden death in AAOCA have been proposed in previously published articles¹⁰: the aortic pressure on an intramural segment, an inter-arterial segment “squeezed” between the aorta and pulmonary

¹ Cardiac Surgery Department, Hospital la Princesa, Madrid, Spain

² Cardiac Surgery Department, Hospital Fundación Jiménez Díaz, Madrid, Spain

³ Cardiology Department, Hospital Fundación Jiménez Díaz, Madrid, Spain

⁴ Radiology Department, Hospital Fundación Jiménez Díaz, Madrid, Spain

Corresponding Author:

Laura Varela Barca, Cardiac Surgery Department, Hospital la Princesa, Madrid, Spain.

Email: lauravarela21089@mail.com

Abbreviations

AAOCA	anomalous aortic origin of a coronary artery
AAOLCA	anomalous aortic origin of the left coronary artery
AAORCA	anomalous aortic origin of the right coronary artery
CABG	coronary artery bypass grafting
IQR	interquartile range
SD	standard deviation

artery, spasm, kinking, or sharp angulation. In addition, the interarterial course of AAOCA has been related to increased risk of ischemia if there is a long intramural segment, acute angulation of the proximal segment or abnormally high take-off from the aorta.^{10,11}

In older adults, AAOCA typically presents with cardiac symptoms such as chest pain or dyspnea. Some authors have reported an association between AAOCA and coronary artery disease,¹² whereas others concluded there is no association.¹³ Some authors have reported that AAOCA contributed to a greater ischemic burden when symptoms appear at younger ages.³

When anatomical restoration treatment is indicated, surgical repair is the gold standard and surgical unroofing is the most common technique. Indications for surgery in AAOCA follow the guidelines of 2017¹⁴ and 2018¹⁵ and, in addition, a systematic review was presented recently by a multidisciplinary group of clinical and interventional cardiologists and cardiac surgeons proposing some recommendations regarding patient management.¹⁶ In general terms, surgery is recommended for individuals with AAOLCA arising from the right sinus of Valsalva and AAORCA arising from the left sinus of Valsalva if there are symptoms, arrhythmias, or ischemia inducible tests are positive. However, in all cases of AAOCA, the decision to perform surgery should be a patient-specific decision.

Although some large patient series have been published in the last few years from American registries, in Europe less data have been published regarding surgical correction of AAOCA. We aim to report our experience with AAOCA surgery and the long-term results.

Patients and Methods

Patients

This is a retrospective observational study including three centers with the same surgical team.

Sixteen adult patients underwent surgical repair of AAOCA in the last ten years, from 2013 to 2023.

The diagnosis was made based on the results of transthoracic echocardiography, computed tomography angiography, and coronary angiography findings.

The coronary morphology was determined based on a combination of imaging studies, operative reports, and clinical notes. Symptoms, comorbidities, and evidence of ischemia came from records closest to the date of the AAOCA diagnosis. Ischemia was defined based on stress test reports of electrocardiography, echocardiography, and myocardial perfusion imaging.

Follow-up was performed using the hospital database and direct contact with all the patients.

Surgical Technique

All operations included were performed by the same surgeon. Surgical unroofing of an intramural AAOCA was the most frequently employed surgical strategy; however, this technique was adapted to the patient's variant of AAOCA. Neo-ostium creation and bypass grafting were also employed.

The three specific techniques employed in AAOCA correction are coronary unroofing, neo-ostium formation, and reimplantation. The decision to perform one of the previously mentioned surgical repairs depends on each patient's characteristics and anatomical aspects, the presence of an intramural segment and a subcommissural portion being the main determinants.

Coronary unroofing is a widely used technique which consists of an incision in the intramural segment from the ostium to the take-off of the anomalous coronary artery. Unroofing avoids significant coronary manipulation and reduces the risk of ischemia. Some of the possible complications of this technique are partial compression of the coronary artery, coronary angulation, residual interarterial segment, or aortic valve injury.^{17–19} Coronary unroofing is the indicated technique in cases of intramural segment without a subcommissural portion.

In an neo-ostium technique, an unroofing of the intramural portion located at the correct anatomic location is performed without an incision of the entire intramural portion.¹⁷ A circumferential suture is placed into the aortocoronary incision to create the neo-ostium. In cases with intramural segment and a subcommissural portion, the neo-ostium is the appropriate technique.

Coronary reimplantation consists of reattaching the anomalous coronary artery to its respective sinus of Valsalva to restore normal physiology. It is the technique employed in cases of no intramural segment.

Statistical Analyses

Descriptive statistics and analyses were conducted using Stata/IC 16 (Stata Statistical Software: Release 146 College Station, TX: StataCorp LP). Results are summarized reporting the absolute numbers and percentages. For categorical variables, mean with standard deviation (SD), when normally distributed, or median with interquartile range (IQR) is given.

Results

Patient Characteristics

Sixteen adult patients underwent surgical correction of AAOCA. Two of the AAOCA were AAOLCA, anomalous origin of the left coronary artery with origin in the right coronary sinus (Figure 1) and 14 were AAORCA, anomalous origin of the right coronary artery from the left coronary sinus (Figure 2).

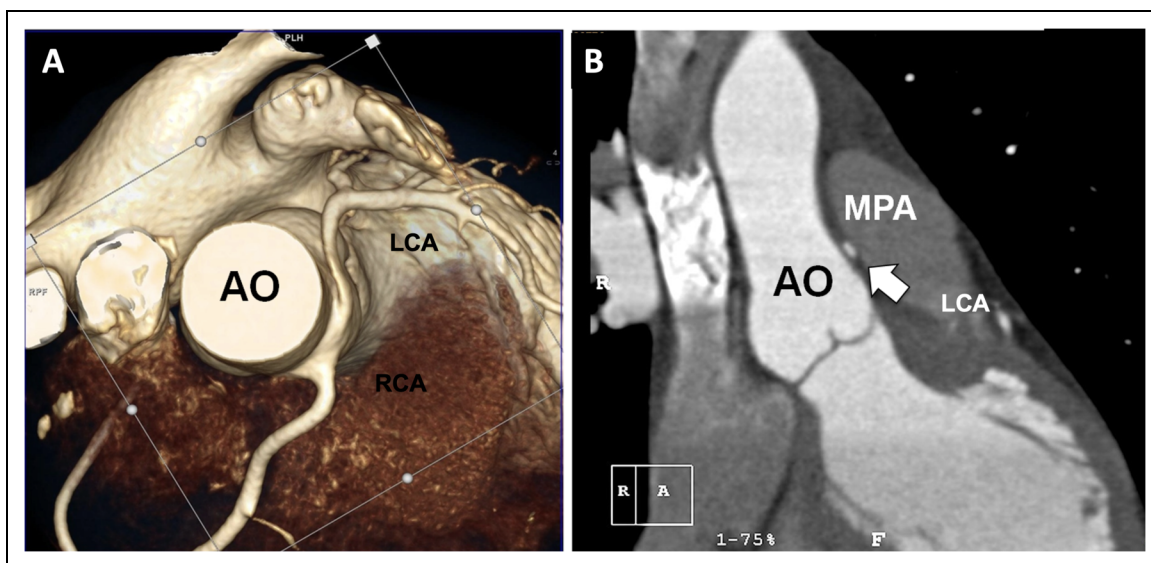


Figure 1. Volume-rendered computed tomography (CT) reconstruction showing the anomalous origin of the left coronary artery from the right coronary sinus of Valsalva (AAOLCA). A: axial plane B: coronal plane. Abbreviations: AO, aorta; LCA, left coronary artery; MPA, main pulmonary artery; RCA, right coronary artery. The arrow shows the LCA as well as its course between the pulmonary artery and the aortic root.

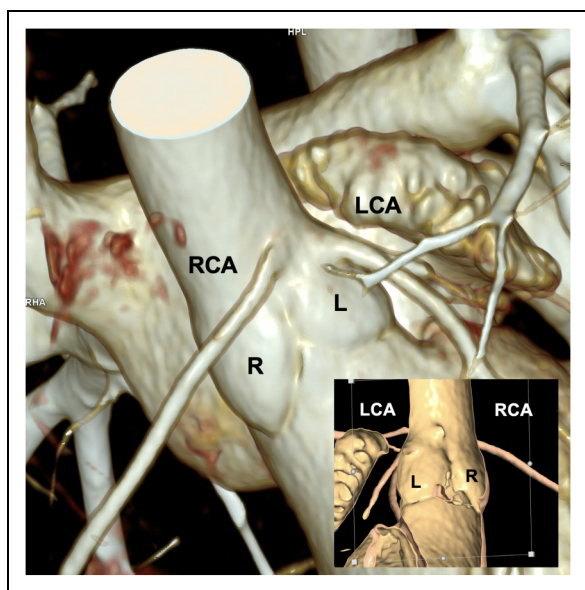


Figure 2. Volume-rendered computed tomography (CT) reconstruction showing the anomalous origin of the right coronary artery (AAORCA) from the left coronary sinus of Valsalva. Endovascular view is shown in the right and lower figure. Abbreviations: L, left sinus of Valsalva; LCA, left coronary artery; R, right sinus of Valsalva; RCA, right coronary artery.

Median age was 43.3 years (IQR 13.9), 8 (50%) patients were males and the remaining eight (50%) were females.

All patients had an intramural segment, defined as a part of the coronary artery path within the aorta²⁰ which can be affected by a fixed-anatomical and dynamic stenosis that can vary during the cardiac cycle causing a systolic lateral compression from the aorta.

Two patients were elite athletes, two were nonelite athletes and the remaining 12 patients did not practice any sport. Table 1 summarizes the information about each patient, AAOCA symptoms, diagnostic test, and surgical technique.

Four patients were asymptomatic when diagnosed (25%) whereas 12 (75%) patients presented with cardiovascular symptoms: 9 (56.3%) suffered from exertional chest pain, syncope was present in 1 (6.25%) patient and 2 (12.5%) patients had aborted sudden cardiac death.

Of the 16 patients, 7 (43.8%) had ischemic manifestations or electrocardiogram changes on a stress test, whereas in 4(25%) patients the test was not performed.

The four asymptomatic patients were diagnosed incidentally after undergoing computed tomography angiography for other clinical reasons. One of them, the 20-year-old elite athlete, underwent a specific test as part of the process to participate in an international sports competition.

Surgical Procedures

Unroofing of an intramural AAOCA was performed in 11 patients. Figure 3 shows a case before and after unroofing of AAORCA, and Figure 4 shows a real image of coronary unroofing.

Three patients underwent neo-ostium creation (Figures 5 and 6). The remaining two patients underwent more complex surgeries. One patient underwent a David procedure with aortic valve sparing and coronary reimplantation; and the other patient was treated by coronary bypass grafting due to coronary atheromatous lesions in the anomalous aortic origin of a coronary artery (Figure 7).

In two patients, a concomitant procedure was performed: replacement of the pulmonary valve due to an associated

Table 1. Baseline Characteristics of Patient.

Id	Sex	Age (years)	Athlete ^a	Symptoms	Ischemia detection test	AAOCA	Surgery
1	Male	47	No	Atypical chest pain	Positive ECG	AAORCA	Unroofing
2	Male	18	Elite	Chest pain during exercising	Negative	AAORCA	Unroofing
3	Male	44	No	Chest pain during exercising	Negative	AAORCA	Neo-ostium
4	Female	35	No	Atypical chest pain	Clinically positive	AAORCA	Unroofing
5	Male	56	No	Dyspnea, chest pain	None performed	AAORCA	Unroofing
6	Male	38	No	No symptoms	None performed	AAORCA	Neo-ostium
7	Female	38	No	Syncope, chest pain	None performed	AAOLCA	Neo-ostium
8	Male	53	No	No symptoms	None performed	AAORCA	Reimplantation
9	Male	20	Elite	No symptoms	MIBI-Spect positive	AAORCA	Unroofing
10	Male	43	Nonelite	Chest pain during exercising	Positive ECG	AAORCA	Unroofing
11	Female	73	No	Sudden cardiac arrest	Negative	AAOLCA	CABG
12	Female	43	Nonelite	Sudden cardiac arrest	MIBI-Spect positive	AAORCA	Unroofing
13	Female	39	No	Chest pain during exercising	Negative	AAORCA	Unroofing
14	Female	47	No	Atypical chest pain	Positive ECG	AAORCA	Unroofing
15	Female	29	No	No symptoms	Negative	AAORCA	Unroofing
16	Female	55	No	Atypical chest pain	Clinically positive	AAORCA	Unroofing

Abbreviations: AAOCA, anomalous aortic origin of a coronary artery; AAOLCA, anomalous aortic origin of the left coronary artery; AAORCA, anomalous aortic origin of the right coronary artery; CABG, coronary artery bypass grafting; ECG, electrocardiography; .

^aAthlete: No athlete, Elite athlete, or Nonelite (athlete but not in an elite competition).

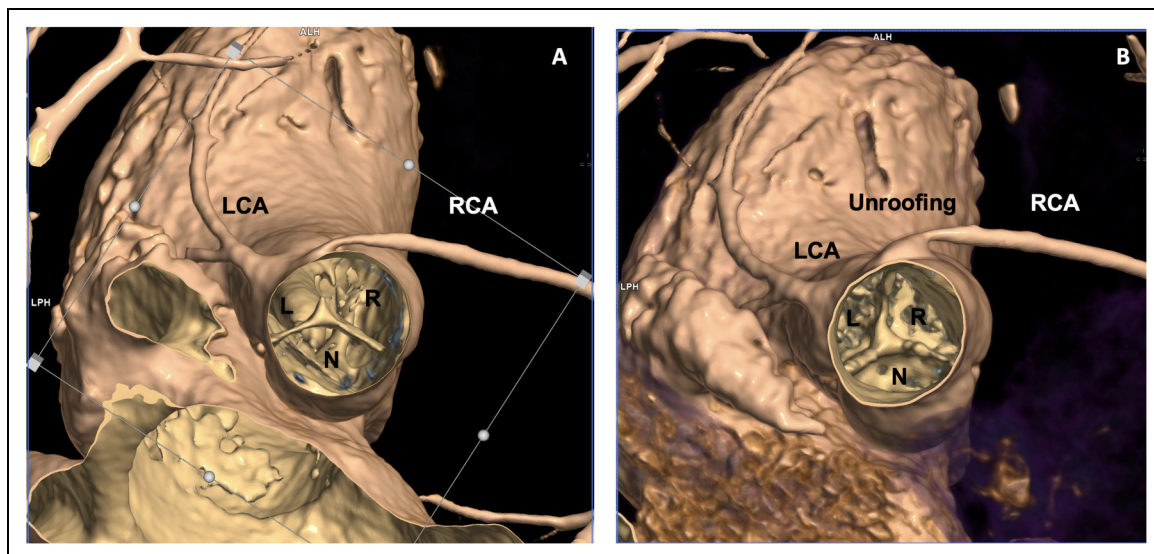


Figure 3. Volume-rendered computed tomography (CT) reconstruction showing the anomalous origin of the right coronary artery from the left coronary sinus of Valsalva (AAORCA). (A) Preoperative reconstruction. (B) Postoperative reconstruction showing the unroofing technique. L, left sinus of Valsalva; LCA, left coronary artery; N, noncoronary sinus of Valsalva; R, right sinus of Valsalva; RCA, right coronary artery;

congenital pulmonary valve stenosis in one patient and an atrial communication closure due to an atrial septal defect in the other.

Outcomes

The mean length of stay was 8.1 days (SD 4.7), median length of stay was 7 days (IQR, 7-8). There was no operative mortality.

All patients were discharged home without complications (neither ischemia or heart failure). Oral treatment with 100 mg

per day of Aspirin was scheduled for three months after the surgical procedure. No exercise restrictions were recommended after the first three months, regarding the recommendations in the Guidelines on Sports, Cardiology and Exercise in Patients with Cardiovascular Disease.²¹

Follow-up period was 73.8 months, there was no late mortality.

In the follow-up echocardiography, none of the patients showed significant aortic regurgitation (considering significant more than mild regurgitation).

Two patients reported chest pain during the follow-up period. In both of them, ischemia detection testing was negative; echocardiography showed no alterations, and coronary angiography was performed showing no lesions and an intact surgical correction. These two patients also suffered from anxiety and nervousness; therefore, an anxiety disorder was

suspected. The chest pain disappeared with anxiety treatment in both cases.

Discussion

AAOCA is a rare congenital lesion with high importance in clinical practice due to the association with myocardial ischemia and sudden cardiac arrest/death in young patients. Although AAOCA was commonly diagnosed in patients with ischemic symptoms, in recent years, it is frequently identified in screening imaging studies in asymptomatic patients.^{2,22} When surgical repair is indicated, the most commonly used technique is surgical unroofing. Large series in the USA have been published showing excellent early outcomes,^{18,23–25} however, in Europe less has been published in reference to surgical approach of AAOCA.^{26,27}

Our study reviewed 16 patients treated with surgical correction of AAOCA who had an excellent outcome. Although it is a small sample of patients with AAOCA, as far as we know, it is the only reported series of cases in which all patients underwent a surgical correction of AAOCA by the same surgeon. In addition, isolated case reports were published in Spain,^{28–30} and a few studies were published in Europe^{26,27} without any follow-up period.

The largest series in Europe,²⁷ published by Padalino et al, reviewed data from 16 cardiac centers in Europe including 156 patients who underwent surgical repair, 22.4% with AAOLCA and 67.3% with AAORCA. The authors reported a low incidence of operative deaths (1.3% due to two emergent cases) and morbidity. However, the authors did not find differences in terms of early survival comparing with patients who had not undergone surgery. Surgery provided benefits in terms of returning to normal activity when comparing patients with medical treatment alone.

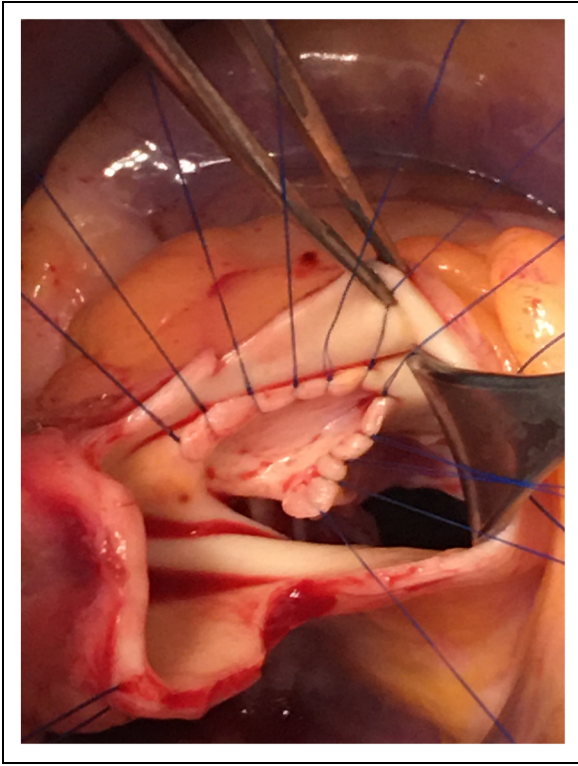


Figure 4. Unroofing technique. Surgical image. Coronary unroofing consists of an incision of the intramural segment from the ostium to the emergence of the anomalous coronary.

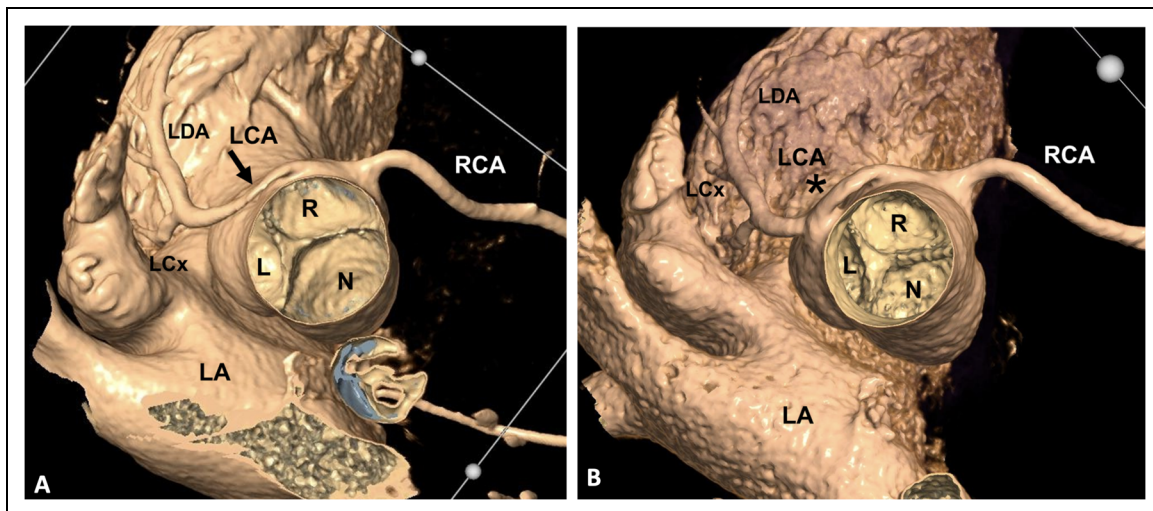


Figure 5. Volume-rendered computed tomography (CT) reconstruction showing the anomalous origin of the left coronary artery from the right coronary sinus of Valsalva (AAOLCA). (A) Preoperative reconstruction. (B) Postoperative reconstruction showing the neo-ostium creation technique (*). L, left sinus of Valsalva; LA, left atrium; LAD, left anterior descending coronary artery; LCA, left coronary artery; LCx, left circumflex coronary artery; RCA, right coronary artery; N, noncoronary sinus of Valsalva; R, right sinus of Valsalva.

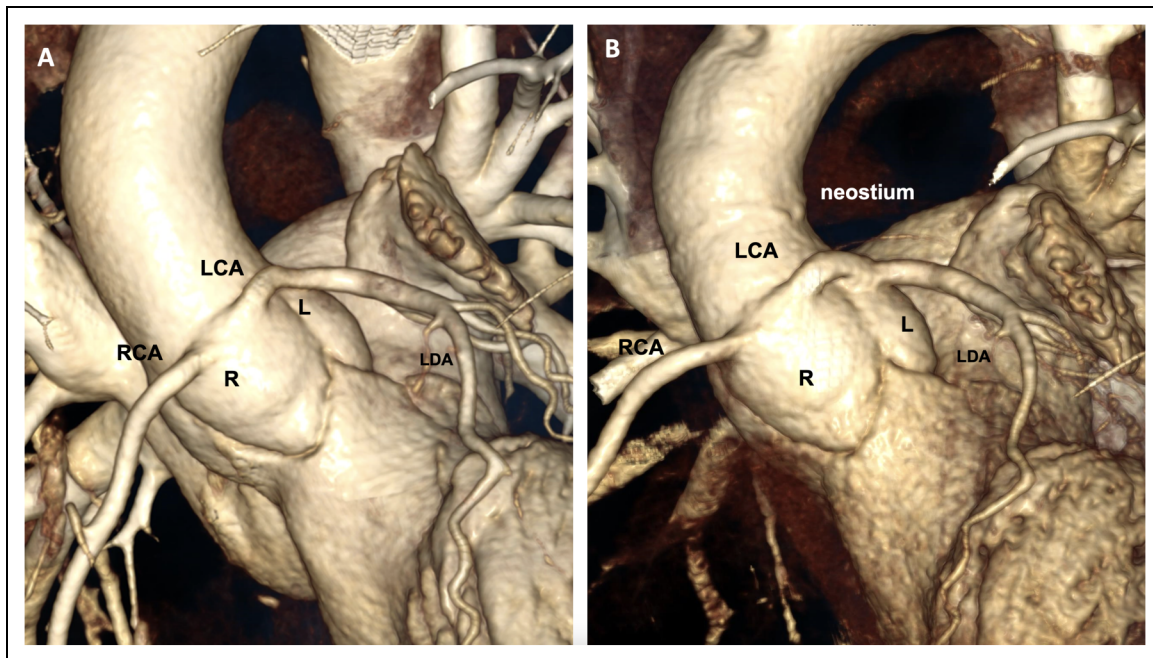


Figure 6. Volume-rendered computed tomography (CT) reconstruction showing the anomalous origin of the left coronary artery from the right coronary sinus of Valsalva (AAOLCA). (A) Preoperative reconstruction. (B) Postoperative reconstruction showing the neo-ostium technique. LCA, common trunk of the left coronary; L, left sinus of Valsalva; LDA, left anterior descending coronary artery; R, right sinus of Valsalva; RCA, right coronary artery; R, right sinus of Valsalva.

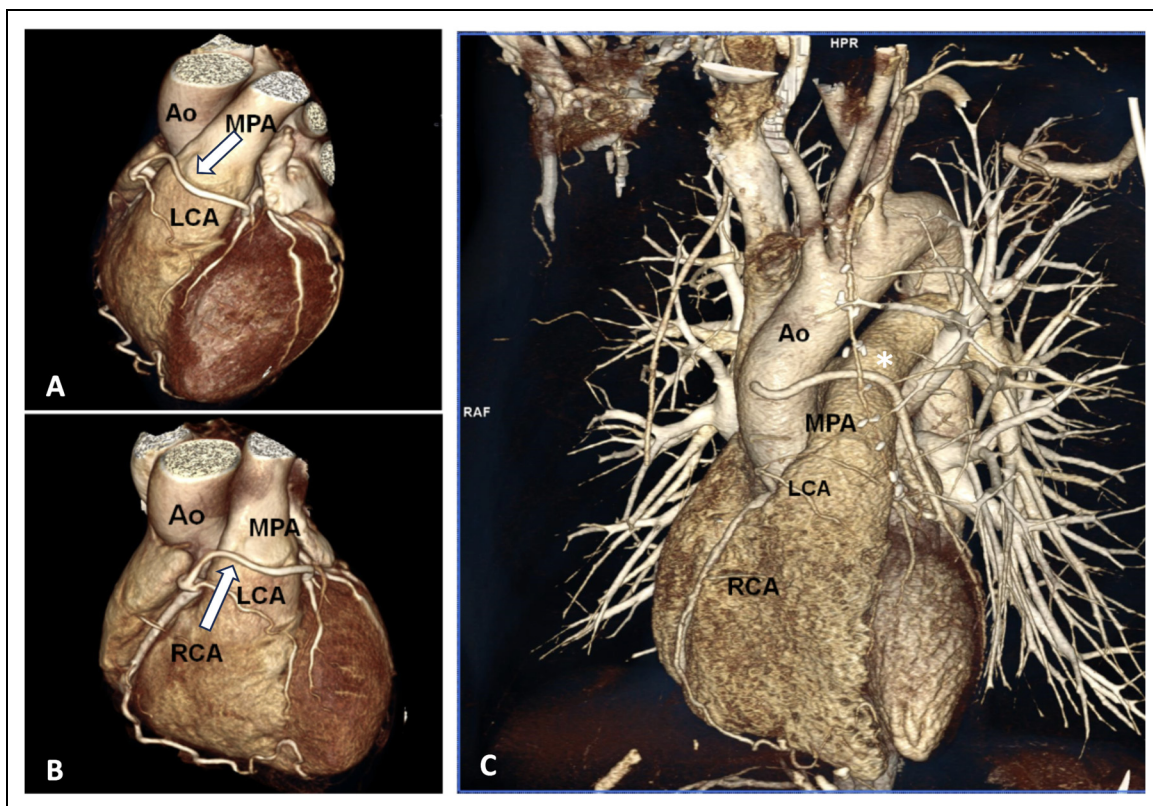


Figure 7. Volume-rendered computed tomography (CT) reconstruction showing anomalous origin of the left coronary artery (AAOLCA) in a prepulmonary course in a patient who was treated by coronary artery bypass grafting (*). Abbreviations: Ao, aorta; LCA, left coronary artery (arrow); MPA, main pulmonary artery; RCA, right coronary artery;

Previously Blomjous et al²⁶ reported 40 cases of AAOCA in adults from which only one underwent surgery. Anomalous origin of the right coronary artery was diagnosed in 90% of the patients and AAOLCA in the remaining 10%. Sudden cardiac death was reported in just one patient, the one who underwent cardiac surgery. The authors were more conservative than other studies as their series showed a better prognosis in adults that did not undergo surgery. Additionally, the authors performed a systematic review of the literature including 20 studies and 1,193 patients. Cardiac surgery was performed in 32.6% of patients. The reported mortality rate was 6.3% during a follow-up of four years, higher than the nonsurgical patients. The systematic review did not demonstrate a protective role of cardiac surgery in AAOCA; however, a long-term follow up was not performed.

In our experience, surgical repair of AAOCA can be performed safely and should be considered after a careful risk assessment. Surgical correction could even be offered to asymptomatic patients if there is a risk of ischemia or sudden cardiac arrest, always taking into account the patient-sharing decision.

Type of Surgical Repair

Unroofing of the intramural segment is the more widely used technique since it avoids significant coronary manipulation.^{6,17} Some limitations described for this technique included that it does not always remove the interarterial segment of the coronary artery, and the risk of aortic regurgitation.^{17,31} However, unroofing has proven to be a feasible and simple surgical technique with excellent outcomes.^{7,23,32}

Other surgical approaches have been proposed. Law et al in a study of 16 cases in Australia³³ reported that surgical reimplantation of the anomalous right coronary artery can be performed safely with excellent medium to long-term outcomes. Bonilla-Ramirez et al¹⁷ in a very good quality review of the different surgical techniques, also compared unroofing, reimplantation, and neo-ostium in AAOCA correction. We believe that the surgical repair technique should be individualized depending on each patient's anatomy, and that the neo-ostium technique has better outcomes than unroofing in cases with a subcommisural course of the AAOCA. Although good results are published regarding all types of surgical approaches, it is important to note the significant technical challenges associated with coronary artery repair.

Unroofing of the coronary artery was the procedure we chose in most of our patients since it is a safe surgical technique that allows a satisfactory surgical correction. However, it is important to adequately match the treatment strategy to each patient to offer the best correction strategy, and for that reason we also performed complex procedures and neo-ostium correction in indicated cases.

Indication of Cardiac Surgery in AAORCA

One interesting point is the surgery indication of AAORCA. Anomalous origin of the right coronary artery was considered

for many years to be a benign type of AAOCA. While AAOLCA was always considered a high-risk entity due to its well-reported association with ischemia and sudden cardiac death, AAORCA seemed to be less associated with symptoms.¹⁸ Anomalous origin of the right coronary artery is six to ten times more common than AAOLCA and, regarding previous articles, it is less associated with cardiac ischemia.³⁴ Brothers et al³⁵ calculated a cumulative risk of death over a 20-year period in athletes to be 6.3% for AAOLCA and 0.2% in AAORCA. Moreover, Jegatheeswaran et al³⁶ in a large cohort of 560 patients found that AAORCA is also associated with ischemia and sudden cardiac death. The authors reported that AAORCA might not always be benign, since it was present in half of the patients with ischemia and one-third of those with sudden cardiac events. These results are consistent with a series from Doan et al of 220 AAORCA patients,³⁷ in which 56 (26%) patients were identified as having high-risk features of sudden death and consequently surgical repair was performed. However, management of asymptomatic AAORCA in a young, active person is challenging,²³ since the surgical intervention has the potential risk of complications.³⁶ The risk of developing aortic regurgitation is about 10% in the published series^{17,36} and up to 20 to 50% of patients reported chest pain after surgery.^{19,36} Cardiac arrest after surgery, although a rare entity, has also been described,³⁸ and restenosis of the coronary ostia and progressive aortic valve regurgitation have also been reported as late complications after surgical unroofing.^{32,39}

We operated on 14 patients diagnosed with AAORCA, 25% asymptomatic, and all of them had an excellent outcome without surgical complication. In our opinion unroofing is a feasible, safe, and reproducible procedure that should be considered in these patients.

Results in Large Series

Recently, the Mayo Clinic²³ published their experience with a protocolized approach to AAOCA treatment in 148 consecutive patients who underwent surgical unroofing; 87.8% of the cases were AAORCA and the remaining 11.5% were AAOLCA. The authors reported a 1% incidence of early deaths, all due to reoperations for aortic root or valve replacement. After a follow-up period, late survival after AAOCA repair at 10 and 15 years was 94.5% and 94.5%. This is the largest series published regarding AAOCA surgery, and the conclusions were that surgical unroofing is a safe procedure with excellent outcomes.

Similarly, Mainwaring et al⁴⁰ published the results regarding AAOCA surgery from Lucile Packard Childrens Hospital in Stanford, California. One hundred and fifteen patients were operated; unroofing was the surgical technique employed in 86 patients, reimplantation in 9, and pulmonary artery translocation in 20, all with excellent outcomes. No early or late mortality were reported by the authors.

Regarding the Cleveland Clinic results,²⁵ 167 patients were diagnosed with AAOCA: 57% involving AAORCA, 23%

AAOLCA, and 19% other type of AAOCA. Only 36 (22%) patients were treated surgically and unroofing was the most employed technique, being used in a higher proportion of AAORCA than AAOLCA; 47% of the patients were treated with unroofing technique, however, within those cases the authors included patients in which a neo-ostium was created. Although the surgical approaches varied, outcomes were favorable for the patients repaired, without any postoperative deaths, and complications were limited to new atrial arrhythmias.

Poynter et al⁷ reported 113 cases of surgical repair for AAOCA in patients younger than 30 years at 29 Congenital Heart Surgeons Society institutions. Unroofing was the surgical technique employed in 88% of the cases.

Conclusions

Although our series has fewer patients when compared with the large series published in the United States, little has been published in Spain and in Europe regarding AAOCA. As we mentioned, Blomjous et al²⁶ have published the largest series in Europe and only one patient underwent cardiac surgery. We operated on 16 patients and all of them had excellent outcomes. In addition, as far as we know, this is the only study published in our continent that includes a long follow-up period.

Our study has the limitations inherent to retrospective studies; however, no patient was missing. This is an observational study at a tertiary center with a selected population of patients and therefore subject to referral bias. Although the follow-up period is longer than several other previously published studies, even longer follow-up is clearly desirable. Similarly, a higher sample size is necessary to obtain more robust results.

Despite all these limitations, our clinical experience has shown that surgical repair of AAOCA can be performed safely without surgical mortality or complications.

In our patients, the surgical repair of AAOCA by unroofing or neo-ostium creation has been shown to have a very good outcome with excellent late survival.

The follow-up period, essential to accurately balance the risks of repairing AAOCA, showed the absence of morbidity and complications.

Surgical procedures for the treatment of AAOCA in are an effective strategy in adult patients with AAOCA.

Author Contributions

All authors have contributed to the article.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

ORCID iD

Laura Varela Barca  <https://orcid.org/0000-0002-4605-8801>

References

- Ogden JA. Congenital anomalies of the coronary arteries. *Am J Cardiol.* 1970;25(4):474-479.
- La Gerche A, Baggish AL, Knuuti J, et al. Cardiac imaging and stress testing asymptomatic athletes to identify those at risk of sudden cardiac death. *JACC Cardiovasc Imaging.* 2013;6(9):993-1007.
- Jiang MX, Brinza EK, Ghobrial J, et al. Coronary artery disease in adults with anomalous aortic origin of a coronary artery. *JTCVS Open.* 2022;10:205-221.
- Cheezum MK, Liberthson RR, Shah NR, et al. Anomalous aortic origin of a coronary artery from the inappropriate Sinus of Valsalva. *J Am Coll Cardiol.* 2017;69(12):1592-1608.
- Dodge-Khatami A, Mavroudis C, Backer CL. Congenital heart surgery nomenclature and database project: anomalies of the coronary arteries. *Ann Thorac Surg.* 2000;69(4 Suppl):S270-S297. doi:10.1016/S0003-4975(99)01248-5
- Ong CS, Cameron DE, Jacobs ML. Surgical management of anomalous coronary arteries. *Ann Cardiothorac Surg.* 2018;7(5):604-610.
- Poynter JA, Williams WG, McIntyre S, Brothers JA, Jacobs ML. Congenital Heart Surgeons Society AAOCA Working Group. Anomalous aortic origin of a coronary artery: a report from the Congenital Heart Surgeons Society Registry. *World J Pediatr Congenit Heart Surg.* 2014;5(1):22-30.
- Maron BJ, Doerer JJ, Haas TS, Tierney DM, Mueller FO. Sudden deaths in young competitive athletes: analysis of 1866 deaths in the United States, 1980–2006. *Circulation.* 2009;119(8):1085–1092.
- Eckart RE, Shry EA, Burke AP, et al. Sudden death in young adults: an autopsy-based series of a population undergoing active surveillance. *J Am Coll Cardiol.* 2011;58(12):1254-1261.
- Jacobs ML. Anomalous aortic origin of a coronary artery: the gaps and the guidelines. *J Thorac Cardiovasc Surg.* 2017;153(6):1462–1465.
- Erez E, Tam VKH, Dublin NA, Stakes J. Anomalous coronary artery with aortic origin and course between the great arteries: improved diagnosis, anatomic findings, and surgical treatment. *Ann Thorac Surg.* 2006;82(3):973-977.
- Sidhu NS, Wander GS, Monga A, Incidence KA. Characteristics and atherosclerotic involvement of coronary artery anomalies in adult population undergoing catheter coronary angiography. *Cardiol Res.* 2019;10(6):358-368.
- Suryanarayana P, Kollampare S, Riaz IB, et al. Presence of anomalous coronary seen on angiogram is not associated with increased risk of significant coronary artery disease. *Int J Angiol.* 2014;23(4):243-246.
- Brothers JA, Frommelt MA, Jaquiss RDB, Myerburg RJ, Fraser CD, Tweddell JS. Expert consensus guidelines: anomalous aortic origin of a coronary artery. *J Thorac Cardiovasc Surg.* 2017;153(6):1440-1457.
- Stout KK, Daniels CJ, Aboulhosn JA, et al. 2018 AHA/ACC guideline for the management of adults with congenital heart disease: executive summary: a report of the American College of Cardiology/American Heart Association task force on clinical practice guidelines. *J Am Coll Cardiol.* 2019;73(12):1494-1563.

16. Gaudino M, Di Franco A, Arbustini E, et al. Management of adults with anomalous aortic origin of the coronary arteries: state-of-the-art review. *Ann Thorac Surg.* 2023;116(6):1124–1141.
17. Bonilla-Ramirez C, Molossi S, Caldarone CA, Binsalamah ZM. Anomalous aortic origin of the coronary arteries—state of the art management and surgical techniques. *Semin Thorac Cardiovasc Surg Pediatr Card Surg Annu.* 2021;24:85–94.
18. Mery CM, De León LE, Molossi S, et al. Outcomes of surgical intervention for anomalous aortic origin of a coronary artery: a large contemporary prospective cohort study. *J Thorac Cardiovasc Surg.* 2018;155(1):305–319.
19. Sachdeva S, Frommelt MA, Mitchell ME, Tweddell JS, Frommelt PC. Surgical unroofing of intramural anomalous aortic origin of a coronary artery in pediatric patients: single-center perspective. *J Thorac Cardiovasc Surg.* 2018;155(4):1760–1768.
20. Formato GM, Agnifili ML, Arzufl L, et al. Morphological changes of anomalous coronary arteries from the aorta during the cardiac cycle assessed by IVUS in resting conditions. *Circ Cardiovasc Interv.* 2023;16(7):e012636. doi:10.1161/CIRCINTERVENTIONS.122.012636
21. Pelliccia A, Sharma S, Gati S, et al. 2020 ESC guidelines on sports cardiology and exercise in patients with cardiovascular disease. *Eur Heart J.* 2021;42(1):17–96.
22. Molossi S, Agrawal H, Mery CM, et al. Outcomes in anomalous aortic origin of a coronary artery following a prospective standardized approach. *Circ Cardiovasc Interv.* 2020;13(2):e008445.
23. Patlolla SH, Stephens EH, Schaff HV, et al. Outcomes of a protocolized approach for surgical unroofing of intramural anomalous aortic origin of coronary artery in children and adults. *J Thorac Cardiovasc Surg.* 2023;165(5):1641–1650.
24. Mainwaring RD, Reddy VM, Reinhartz O, et al. Anomalous aortic origin of a coronary artery: medium-term results after surgical repair in 50 patients. *Ann Thorac Surg.* 2011;92(2):691–697.
25. Jiang MX, Blackstone EH, Karamlou T, et al. Anomalous aortic origin of a coronary artery in adults. *Ann Thorac Surg.* 2021;112(4):1299–1305.
26. Blomjous MSH, Budde RPJ, Bekker MWA, et al. Clinical outcome of anomalous coronary artery with interarterial course in adults: single-center experience combined with a systematic review. *Int J Cardiol.* 2021;335:32–39.
27. Padalino MA, Franchetti N, Sarris GE, et al. Anomalous aortic origin of coronary arteries: early results on clinical management from an international multicenter study. *Int J Cardiol.* 2019;291:189–193.
28. Barriales-Villa R, Moris de la Tassa C. Anomalías congénitas de las arterias coronarias con origen en el seno de Valsalva contralateral: ¿qué actitud se debe seguir? *Rev Esp Cardiol.* 2006;59(4):360–370.
29. Cardenal Piris RM, Diaz Fernández JF, Sánchez González C, Gómez Menchero A. Anomalías congénitas infrecuentes en el origen de las arterias coronarias. *CardiCore.* 2010;45(3):123–126.
30. González CT, Lledó AG. Anomalías de la anatomía coronaria. A propósito de un caso.: presentación de un caso y revisión bibliográfica. *Rev Investig Educ En Cienc Salud.* 2023;8(1):113–119.
31. Cubero A, Crespo A, Hamzeh G, Cortes A, Rivas D, Aramendi JJ. Anomalous origin of right coronary artery from left coronary Sinus-13 cases treated with the reimplantation technique. *World J Pediatr Congenit Heart Surg.* 2017;8(3):315–320.
32. Nees SN, Flyer JN, Chelliah A, et al. Patients with anomalous aortic origin of the coronary artery remain at risk after surgical repair. *J Thorac Cardiovasc Surg.* 2018;155(6):2554–2564.e3.
33. Law T, Dunne B, Stamp N, Ho KM, Andrews D. Surgical results and outcomes after reimplantation for the management of anomalous aortic origin of the right coronary artery. *Ann Thorac Surg.* 2016;102(1):192–198.
34. Naimo PS, Buratto E, Konstantinov IE. Commentary: surgery for an anomalous aortic origin of a coronary artery: to do, or not to do? That is the question. *J Thorac Cardiovasc Surg.* 2020;160(3):772–773.
35. Brothers J, Carter C, McBride M, Spray T, Paridon S. Anomalous left coronary artery origin from the opposite sinus of Valsalva: evidence of intermittent ischemia. *J Thorac Cardiovasc Surg.* 2010;140(2):e27–e29.
36. Jegatheeswaran A, Devlin PJ, McCrindle BW, et al. Features associated with myocardial ischemia in anomalous aortic origin of a coronary artery: a Congenital Heart Surgeons' Society study. *J Thorac Cardiovasc Surg.* 2019;158(3):822–834.e3.
37. Doan TT, Sachdeva S, Bonilla-Ramirez C, et al. Ischemia in anomalous aortic origin of a right coronary artery: large pediatric cohort medium-term outcomes. *Circ Cardiovasc Interv.* 2023;16(4):e012631. doi:10.1161/CIRCINTERVENTIONS.122.012631
38. Agrawal H, Sexson-Tejtel SK, Qureshi AM, et al. Aborted sudden cardiac death after unroofing of anomalous left coronary artery. *Ann Thorac Surg.* 2017;104(3):e265–e267.
39. Fabozzo A, DiOrio M, Newburger JW, et al. Anomalous aortic origin of coronary arteries: a single-center experience. *Semin Thorac Cardiovasc Surg.* 2016;28(4):791–800.
40. Mainwaring RD, Murphy DJ, Rogers IS, et al. Surgical repair of 115 patients with anomalous aortic origin of a coronary artery from a single institution. *World J Pediatr Congenit Heart Surg.* 2016;7(3):353–359.